ARC 4443 - Acoustical, Electrical, and Illumination Systems

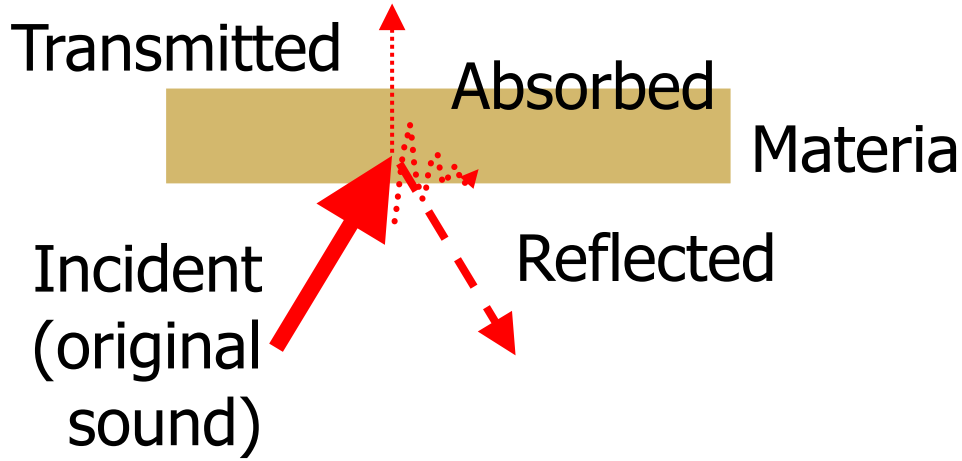
Week 13

Absorption

Spring 2021

Kurt Neiswender, AIA
College of Architecture and Design
Lawrence Technological University

BOUNDARY PHENOMENA



- Incident sound on a boundary (such as a room surface) can be:
  - Reflected (either an echo or diffused sound)
  - Transmitted
  - Absorbed as heat within the material
    - Very small – consider that a 95 dB sound power wave (loud!) contains only 0.0032 watts
    - Neglect heat effect in calculations
- *Most often, all three occur*

Boundary Phenomena

- *Sound absorbing materials*, aka, *acoustical materials* are materials used to reduce *background sound levels* (sound build-up) and *to control reverberation* (sound lingering)
- Sound absorbing materials in the architectural world typically fall into one of the following categories.
  1. **Porous absorbers**
  2. **Panel or membrane absorbers**
  3. **Volume absorbers**

Absorption - General Trends

For all types of sound absorbing materials:

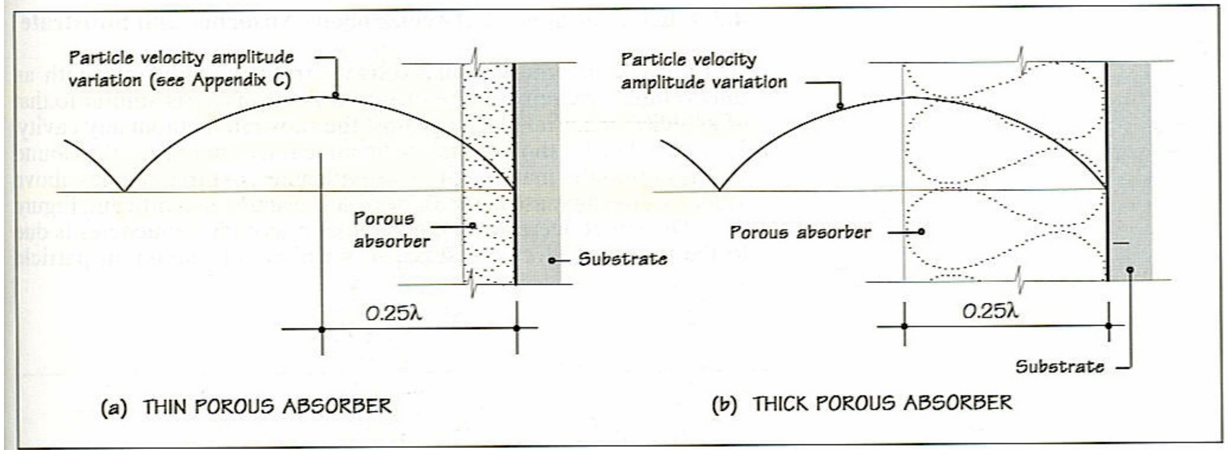
- Absorption is typically greater at high frequencies than low frequencies
- How the material is mounted affects its absorption qualities
- Thicker absorbing materials absorb better at low frequencies, however other attributes also affect their effectiveness such as flow resistance

Absorber Thickness

How do physical dimensions of the porous absorber affect its absorption coefficient (α)?

Absorber Material Thickness: The absorption coefficient at a particular frequency increases with thickness

Frequency (Hz)	Wavelength (mm)	Wavelength (inches)
125	2744	108.3
400	858	33.8
1000	343	13.5
5000	69	2.7
8000	43	1.7

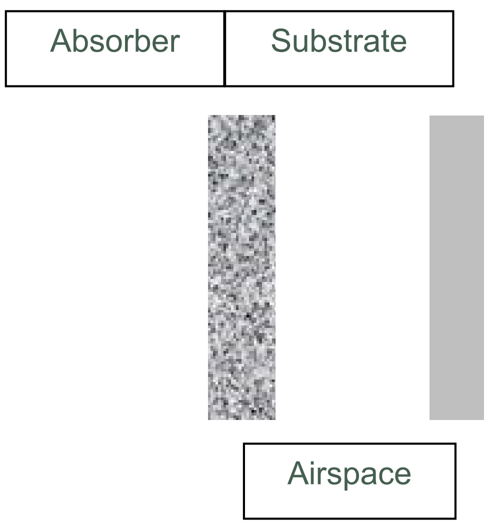


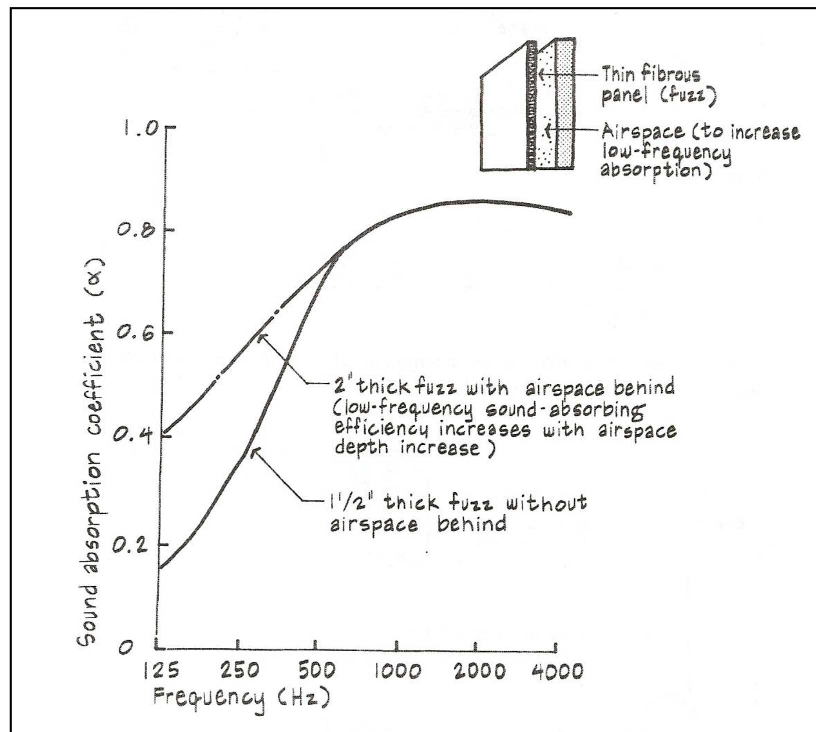
4.2 *Particle velocity amplitude variation superimposed on a section through a wall covered with a porous absorber.*

Absorber Spacing

How does the absorber's spacing from the substrate affect its absorption coefficient (α)?

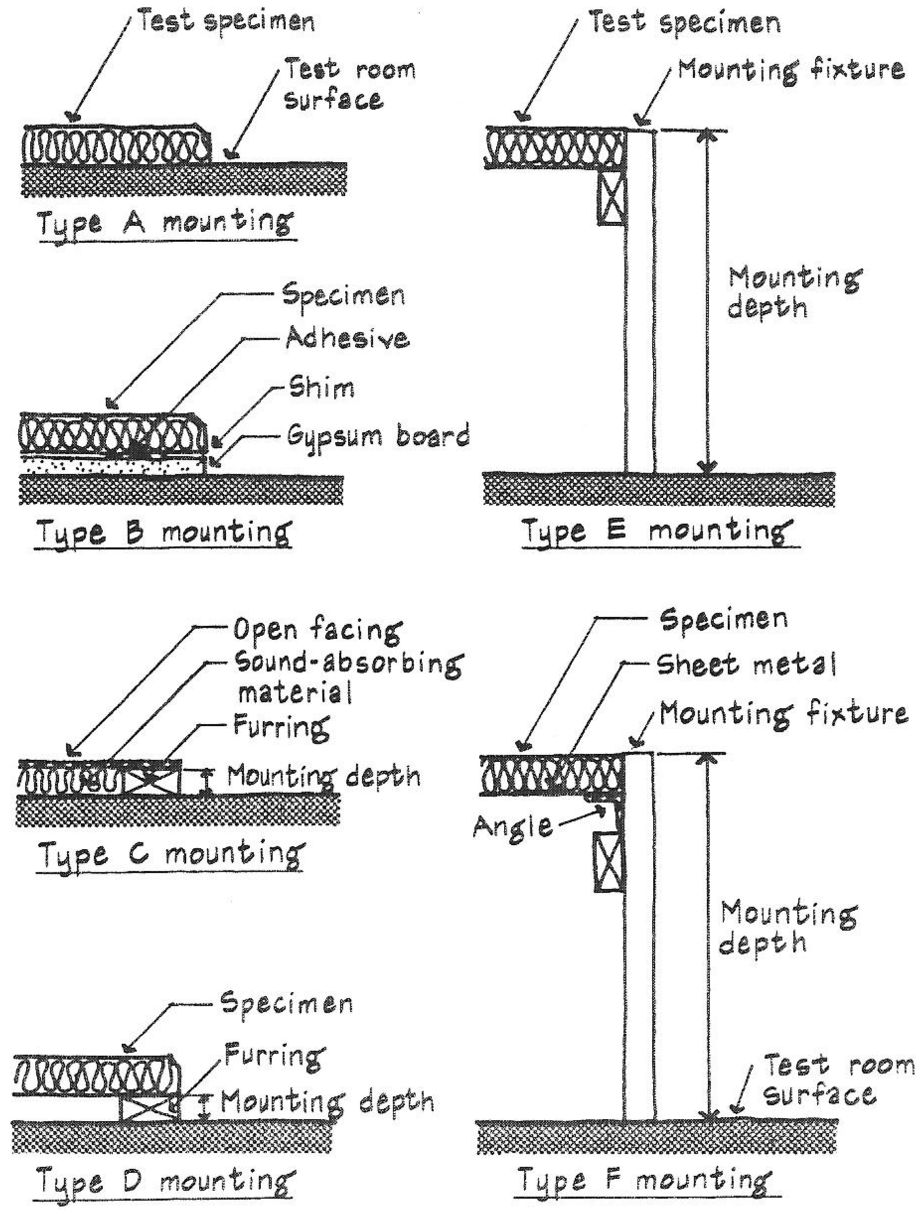
- Increases the absorption coefficient at low frequencies, while not affecting high frequencies where the absorption is already high





Mounting Conditions

- The way that an absorber is mounted affects its absorption coefficient
  - Mounting types are specified in ASTM Std. 795
  - Type A (surface mounted) and type E-400 (400 mm spacing from reflecting surface) mountings will be encountered most frequently. Some materials are tested both ways.



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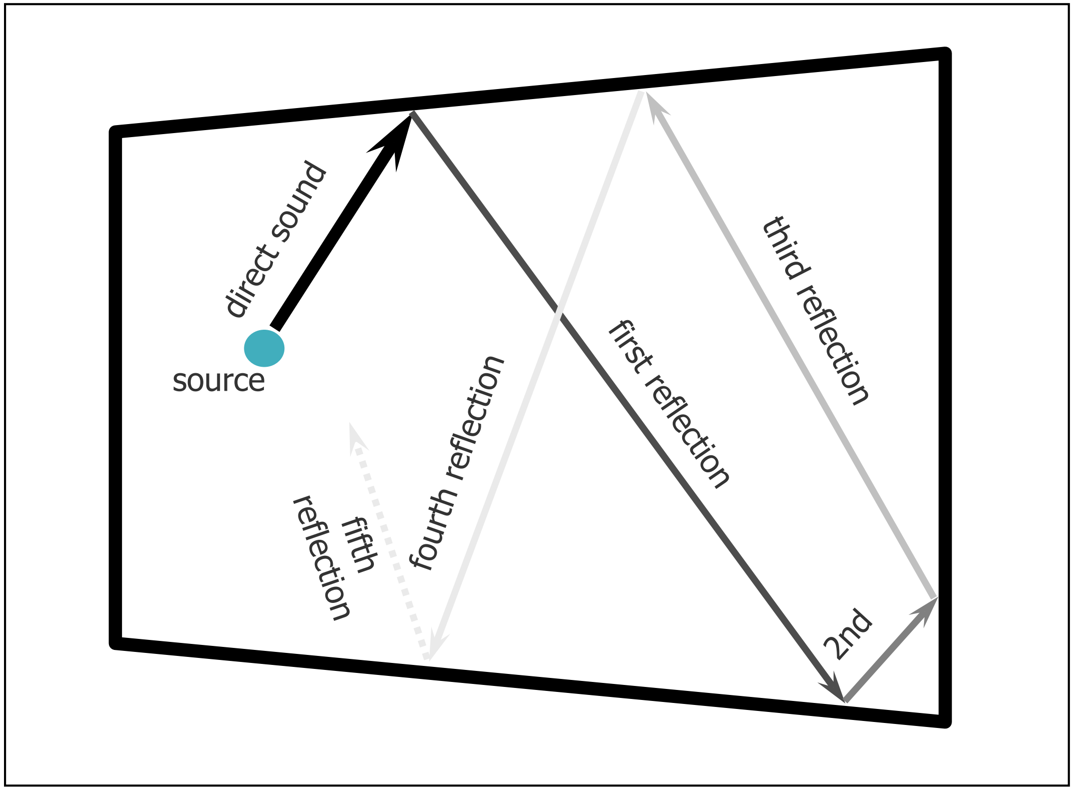
Reverberation

Reverberation

- Sound produced inside a room is contained within that room and bounces off the surfaces until its energy decays completely
- Characteristics of the room influence the sound quality and the sound level
  - room geometry
  - room volume
  - sound absorption, reflection and transmission of room surfaces
  - sound absorption and reflection of room contents

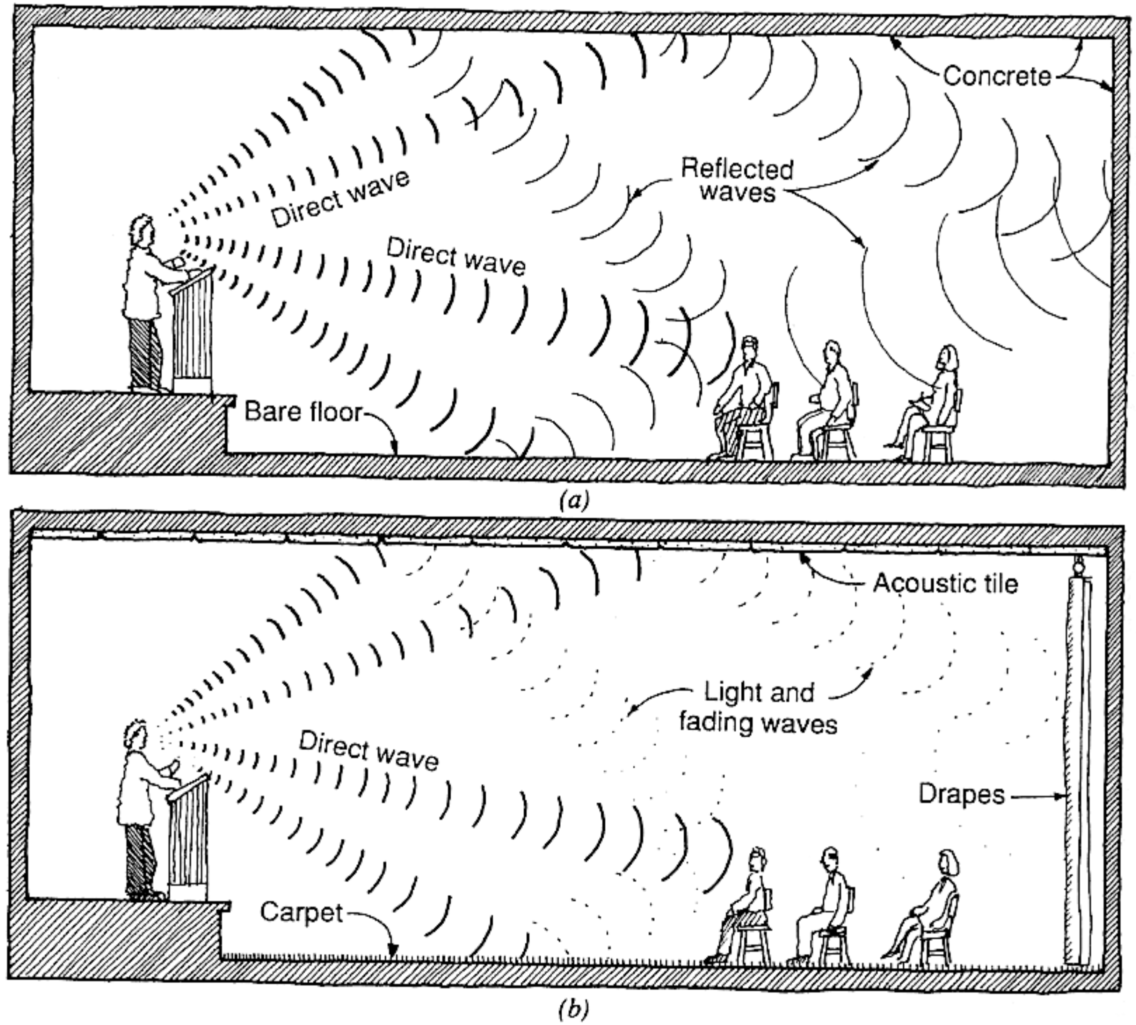
Reverberation

- Reverberation is the persistence of sound in a room after the sources has been terminated.



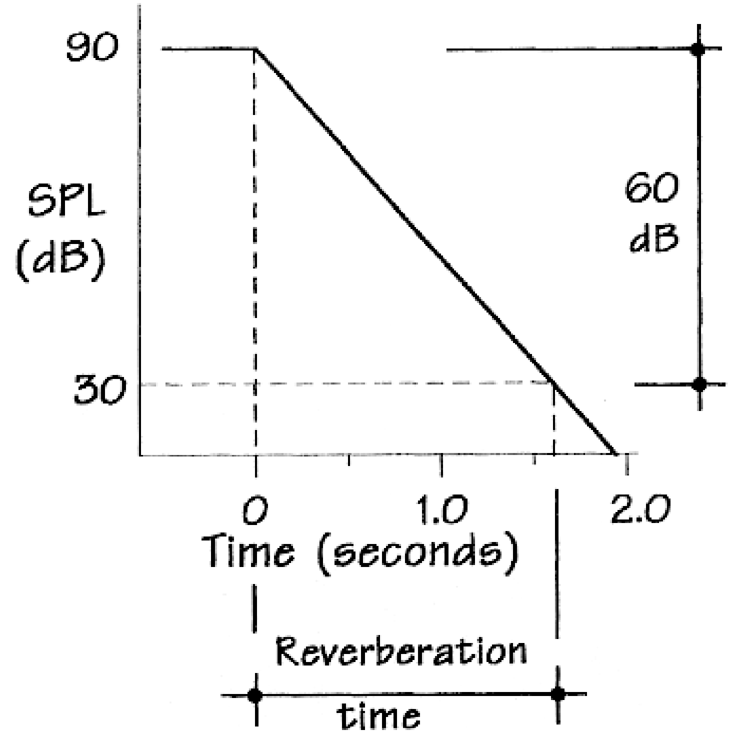
Reverberation

- Of course, sound radiates from a source in many directions



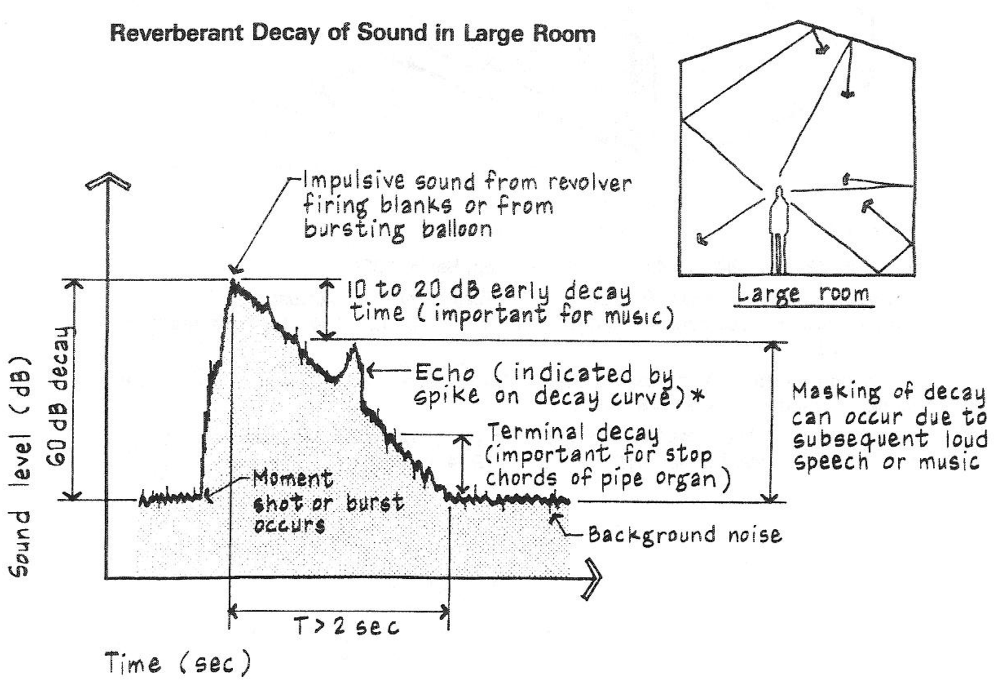
Reverberation Time

- Many sounds are *transient*, or time varying, in nature – ex. music, speech
- The rate of sound decay is called the *reverberation time* (RT), which is *the length of time it takes for a sound to decay 60 dB*
- Reverberant decay graph shows SPL vs. time



Sound Decay

- Measured with a sound level meter or sound analyzer with reverberation time capability
- Can use loudspeaker or impulsive sound (gun blank, large balloon) for sound source



Sound Decay with Echoes

Differences between reverberation and echoes:

- Reverberation is a smooth decrease in the energy of successive reflections.
- An echo is a relatively strong reflection that disrupts the smooth energy decrease.
- Reverberation beneficially reinforces the direct sound
- Echoes are annoying and create unfavorable acoustics

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Absorber types & Applications

Types of Sound Absorbers

- *Sound absorbing materials* are materials used to reduce sound levels and to control reverberation

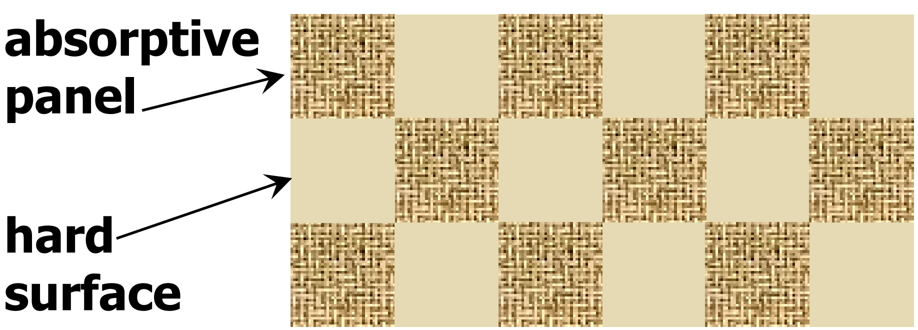
- Three types of sound absorbers
  1. Porous absorbers
  2. Panel or membrane absorbers
  3. Volume absorbers

Sound Absorption Materials: Porous

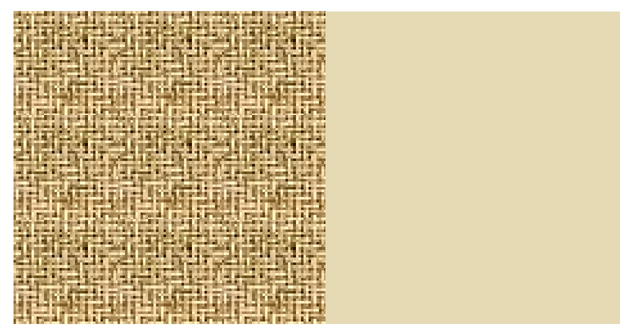
- Fibrous materials work well as porous sound absorbers because of the small air gaps in the material. *Examples are fiberglass and mineral wool → **Typical Insulation Material***
- Generally perform well in the speech range (250-4000 Hz)
- Porous sound absorbing materials which are cellular must have "interconnected cell structures." Examples are polyurethane, melamine and polyester foams.
- Expanded and extruded polystyrene, and polyisocyanurate have closed cells which are <u>great for thermal insulation</u> but <u>poor for sound absorption</u>.

Sound Absorption Materials: Porous

- The efficiency of absorptive panels can be increased by positioning them such that the edges are exposed
  - **Exposed edges absorb sound** – lab test does not include edge area
  - **Diffraction of sound around panel edges “bends” sound toward panel** – more edges equals more diffraction
  - **Also produces some sound diffusion** (scattering)



Pattern with higher absorption



Pattern with lower absorption

Sound Absorption Materials: Porous

- Acoustically transparent – refers to a fabric, screen, perforated surface or other facing that allows sound to easily pass through
- Note: A porous absorber *cannot* be roll or brush painted because this seals the pores. However, a special spraying technique (dryfall) can be used for some materials, such as a factory applied paint to fiberglass panels or field sprayed perforated metal

Porous Sound Absorption Materials: Application – Surface Mounted Panels

Wall or ceiling mounted panels (fiberglass)

acoustical wall panels at
Ferris State University Granger Center

acoustical ceiling panels on
domed ceiling at
Michigan Supreme Courthouse

Porous Sound Absorption Materials: Application – Surface Mounted Panels

Wall or ceiling mounted panels (fiberglass)

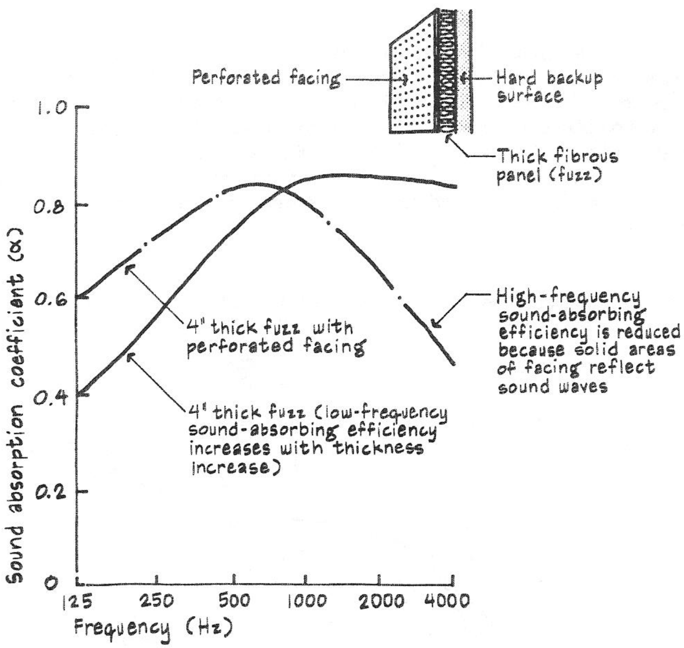
decoustics, Inc. installation – ink on acoustically transparent vinyl over fiberglass (ceiling treatment)

Porous Sound Absorption Materials: Application – Surface Mounted Panels

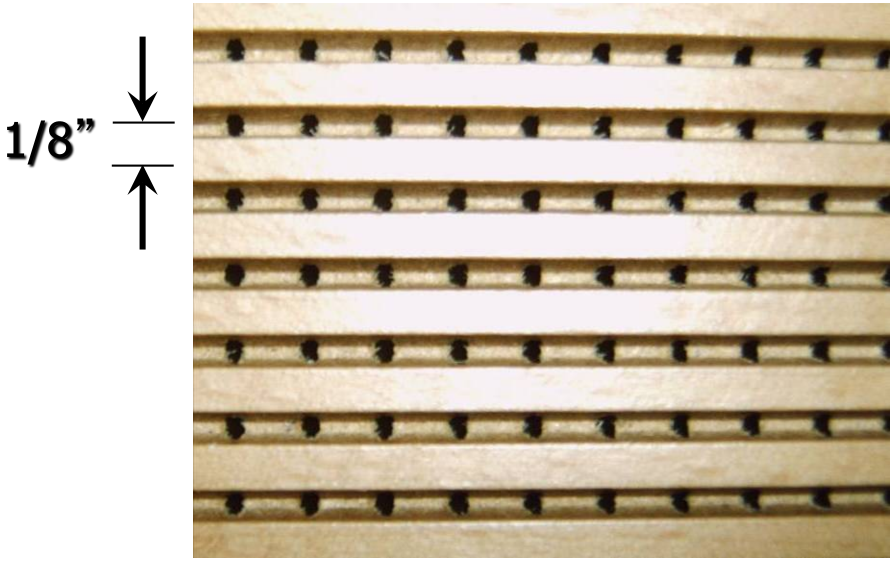
- Wall or ceiling mounted panels
  - Standard thicknesses range from ½” to 4” – most often have 1” or 2” on projects, 4” for music
  - Typically 6-7 pcf density fiberglass core with fabric or perforated vinyl covering
  - “Impact resistant” panels available with 18-20 pcf density (compressed) fiberglass facing on 6-7 pcf fiberglass
  - Can be perforated mineral fiberboard tackable panels, but NRC drops significantly
  - Manufacturer offers fabric or vinyl coverings - Project designer selects

Porous Sound Absorption Materials: Application – Surface Mounted Panels

Rigid screen backed with batt or semi-rigid fiberglass



decoustics Quadrillo® ceiling

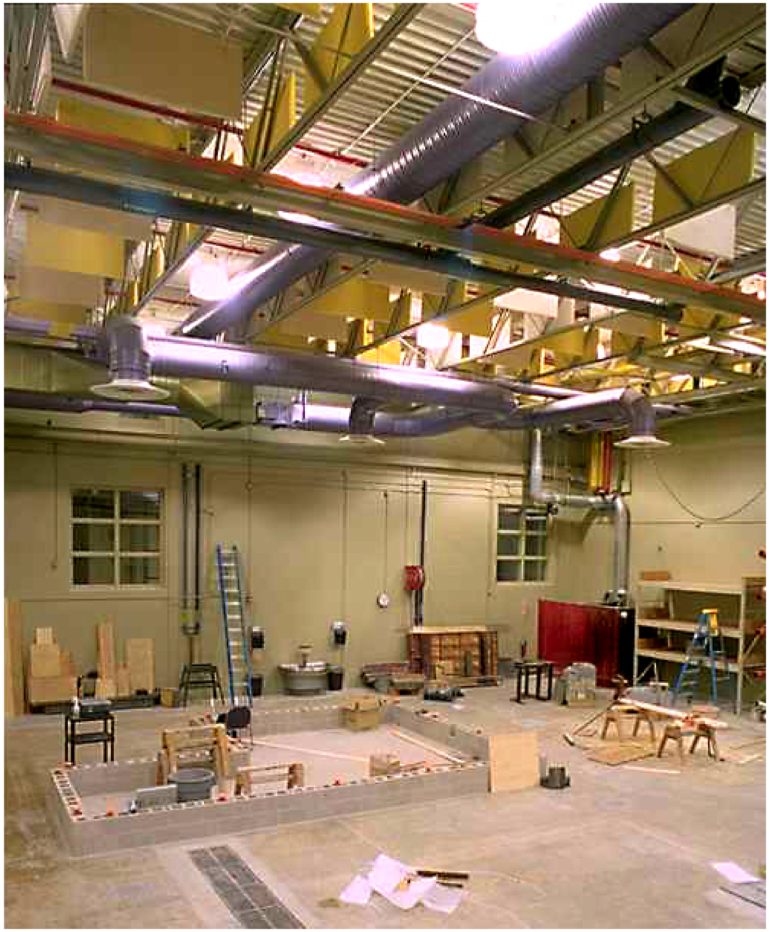


close up of perforations

Porous Sound Absorption Materials: Application – Suspended Absorbers

Flat & dimensional suspended baffles, clouds and banners

Baffles, clouds and banners are typically used with exposed ceilings or where the ceiling is high



baffles at Ferris State University Granger Center

Porous Sound Absorption Materials: Application – Suspended Absorbers

banners: MBI Lapendary® Panels

clouds: Golterman & Sabo (doesn't have to be in "cloud" shape to be an acoustical cloud

Porous Sound Absorption Materials: Application – Suspended Absorbers

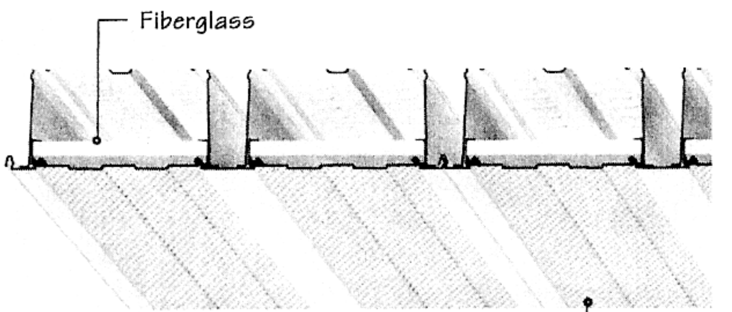
- Space absorbers – any shape is possible, but sound absorption is not known unless tested

RPG Diffusor Systems – Modex CornerTM bass trap

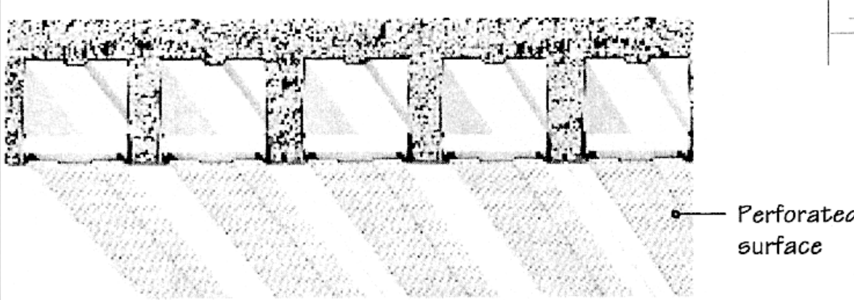
Studio Traps – Silent Source

Porous Sound Absorption Materials: Application – Suspended Absorbers

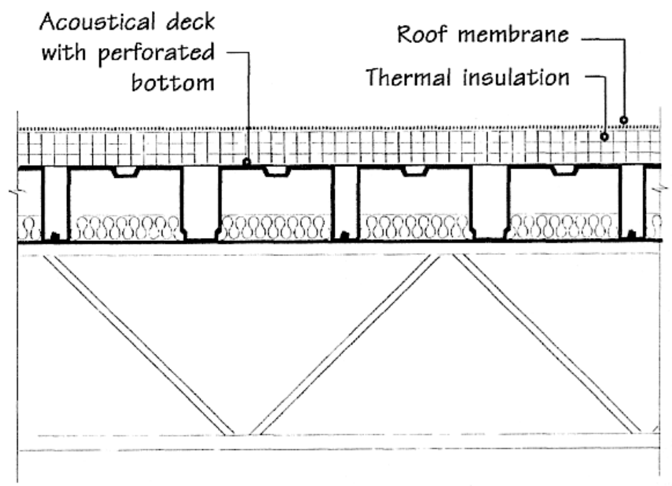
- Acoustical metal deck (floor and roof deck)
  - Perforated metal allows sound to pass into the fiberglass in the voids



ACOUSTICAL ROOF DECK



ACOUSTICAL FLOOR DECK

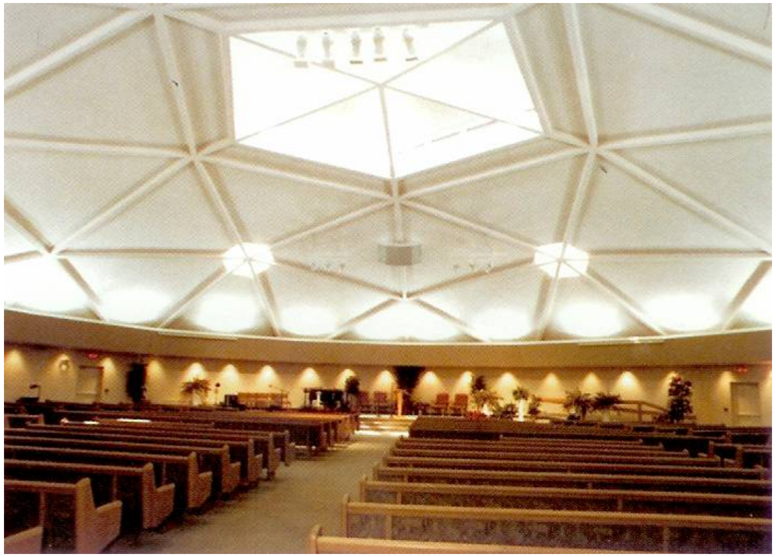


ACOUSTICAL ROOF DECK AND ITS SUPPORT STRUCTURE

Porous Sound Absorption Materials: Application – Suspended Absorbers

- Spray-on acoustical finish – economical for large areas

International Cellulose Corporation
k-13 Spray-on Systems®



International Cellulose Corporation
SonaSpray® "fc"

Porous Sound Absorption Materials: Application – Suspended Absorbers

- Office partitions (office cube matrices) – can be purchased with sound absorptive surfaces and acoustically transparent fabrics

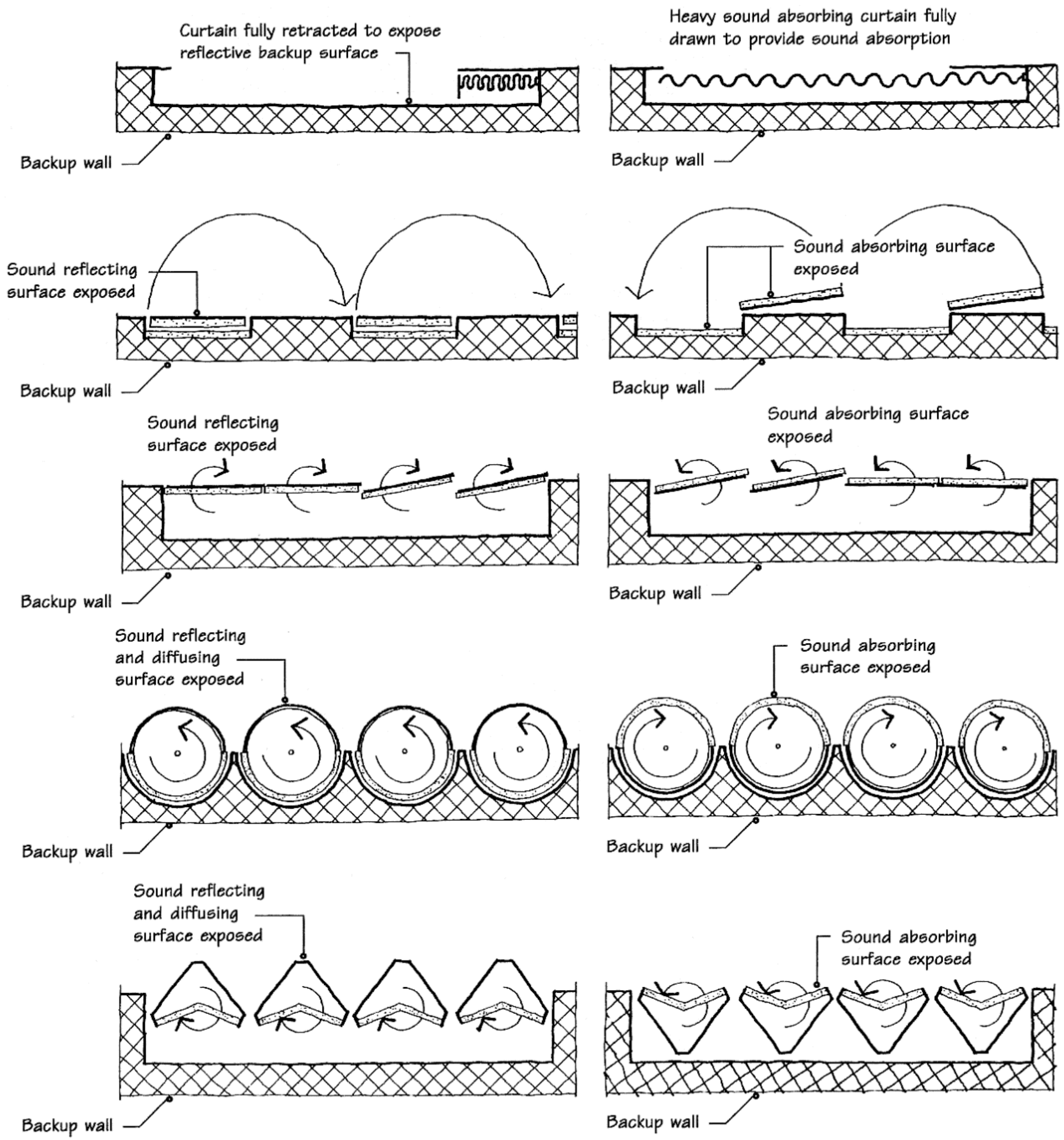
Steelcase, Inc.

Porous Sound Absorption Materials: Application – Misc.

- Acoustical plaster
  - "Seamless" monolithic appearance
  - Can be applied to irregularly shaped surfaces
  - Example 1 - Gypsum based
    - A lightweight porous or fibrous matrix and binder
  - Example 2 - acoustically transparent emulsion that is trowel applied over fiberglass panels
- Draperies
  - Must be heavy weight fabric (20 oz/yd or more)
  - Most often used for variable absorption in halls, theaters and music practice rooms
- Open window - α=1.0 because no sound bounces back
- Open door to corridor – most sound doesn't bounce back

Variable Absorption

Variable sound absorption allows for surfaces to be altered from reflective, to absorptive to diffusive to adjust for speech or music use.



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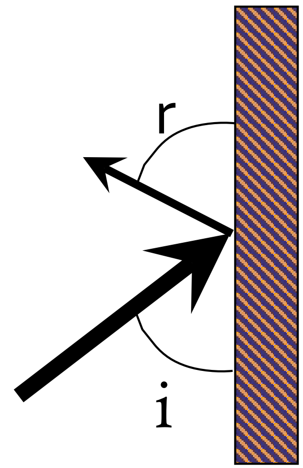
Sound Reflection & Diffusion

Sound Reflection

- Reflection – sound that is reflected back into the space
  - Useful to reinforce or enhance sound



$\angle i = \angle r$

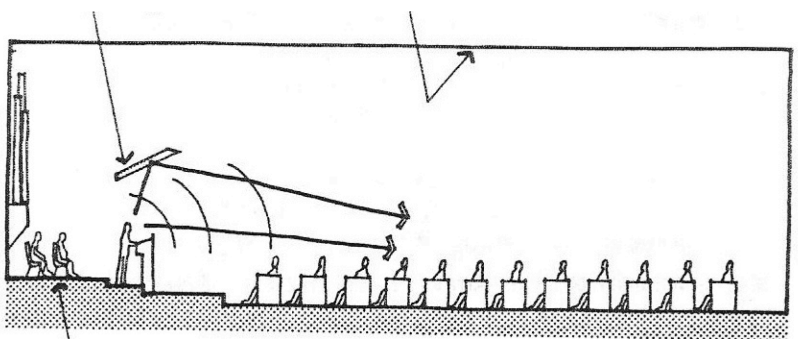
Angle of incidence
equals angle of refection

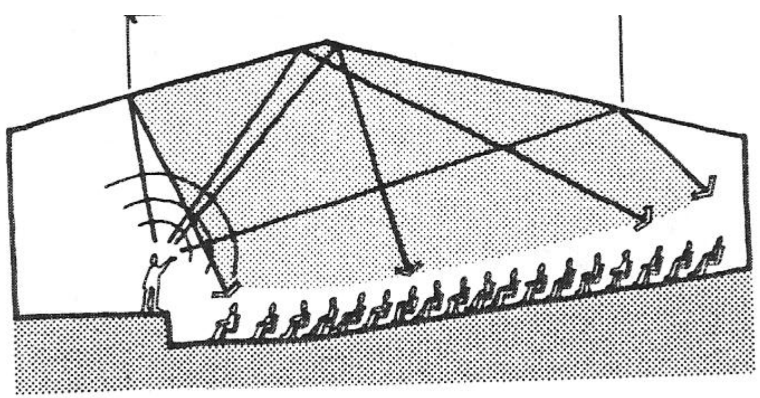
Sound Reflection

Useful reflections direct sound to audience

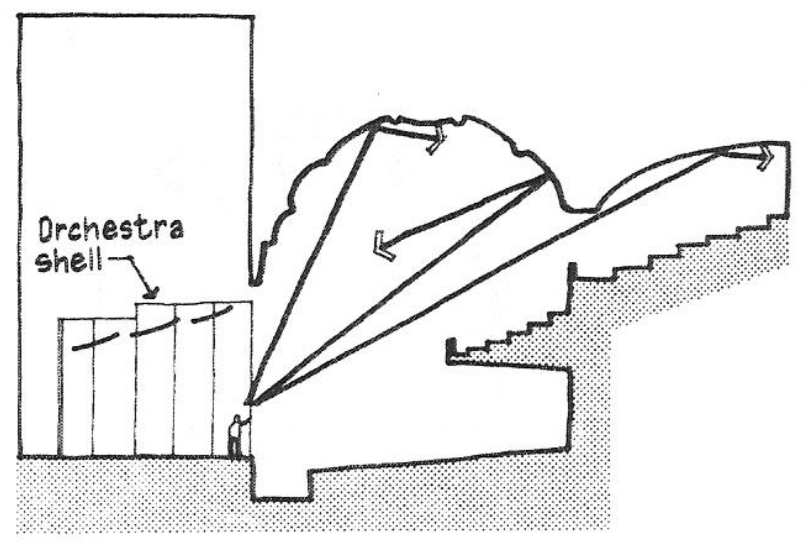
- ray diagram

Useful reflections





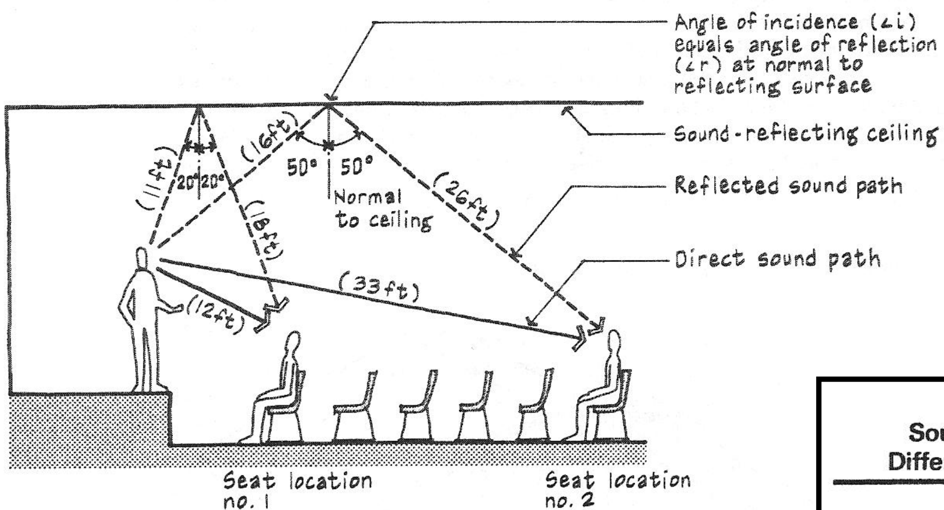
Reflections that are not useful

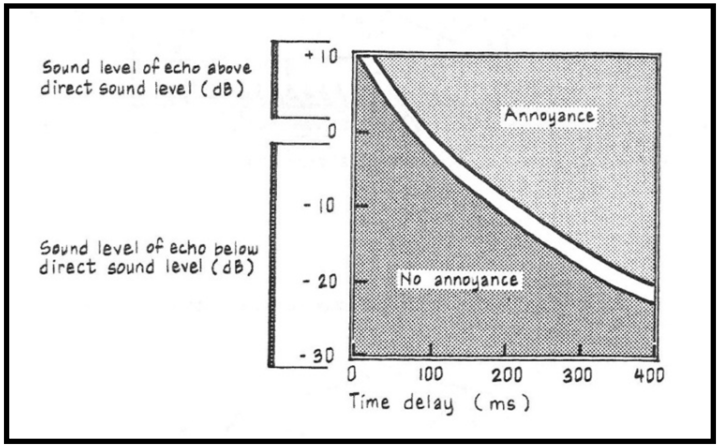


Sound Reflection

Echo - specular reflection that arrives 55-60 milliseconds or more delayed from the direct sound at a sufficient enough sound level above background to be heard as distinct

- sound path difference = reflected path – direct path (ft or m)
- time delay (in ms) = 1000*(sound path difference/c)

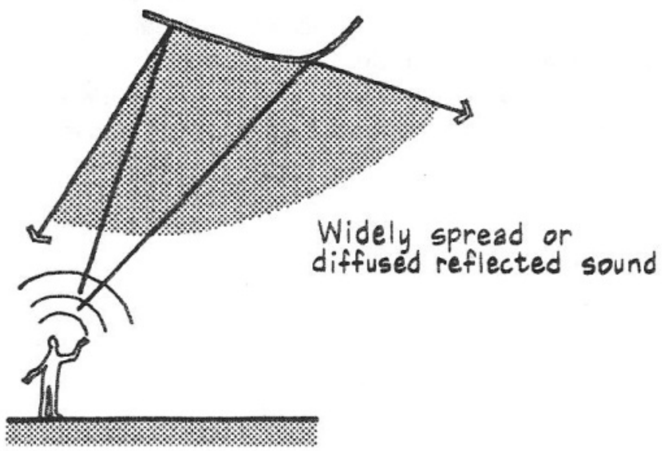


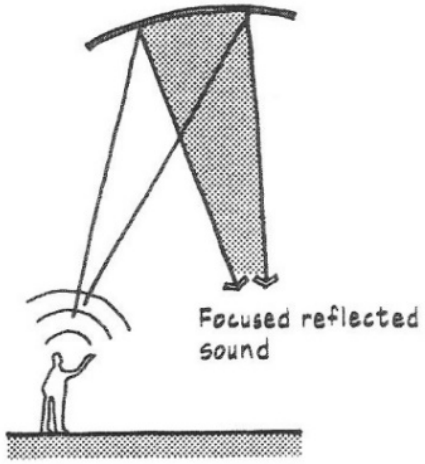


Sound Path Difference (ft)	Time Delay Gap (ms)	Listening Conditions
< 23	< 20	Excellent for speech and music
23 to 34	20 to 30	Good for speech, fair for music
34 to 50	30 to 45	Marginal (*blurred*)
50 to 68	45 to 60	Unsatisfactory
> 68	> 60	Poor (*echo* if strong enough)

Sound Reflection

Convex and Concave Surfaces





convex reflector
sound scattering

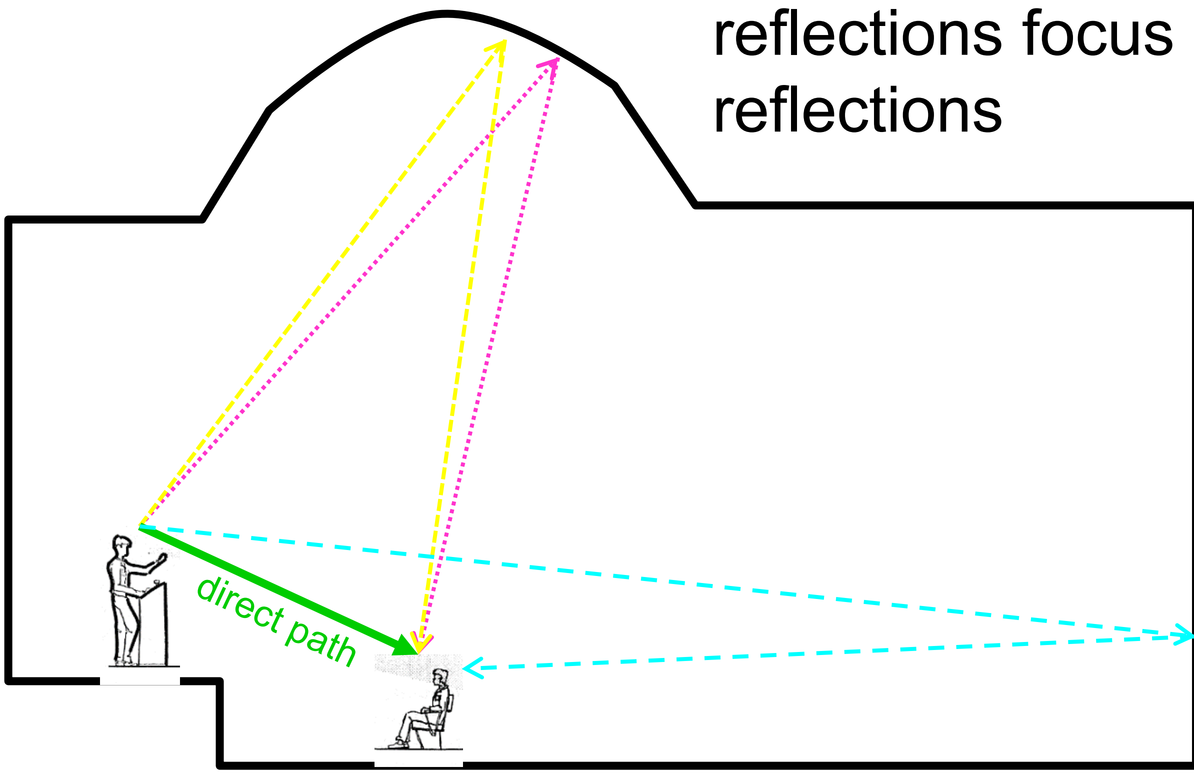
concave reflector
sound focusing

- Convex surfaces *increase diffusion by scattering sound*
- Concave surfaces tend to *focus or concentrate sound into one direction and location*

Echo Scenario

Large differences (>68 ft) between short direct path and long reflection paths can create echoes

concave (e.g. dome) surface reflections focus and amplify reflections

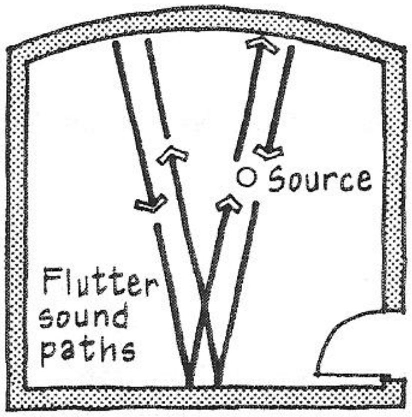


Back wall reflection

Sound Reflection

Flutter Echo - rapid and repetitive sound reflections between hard, flat reflecting surfaces

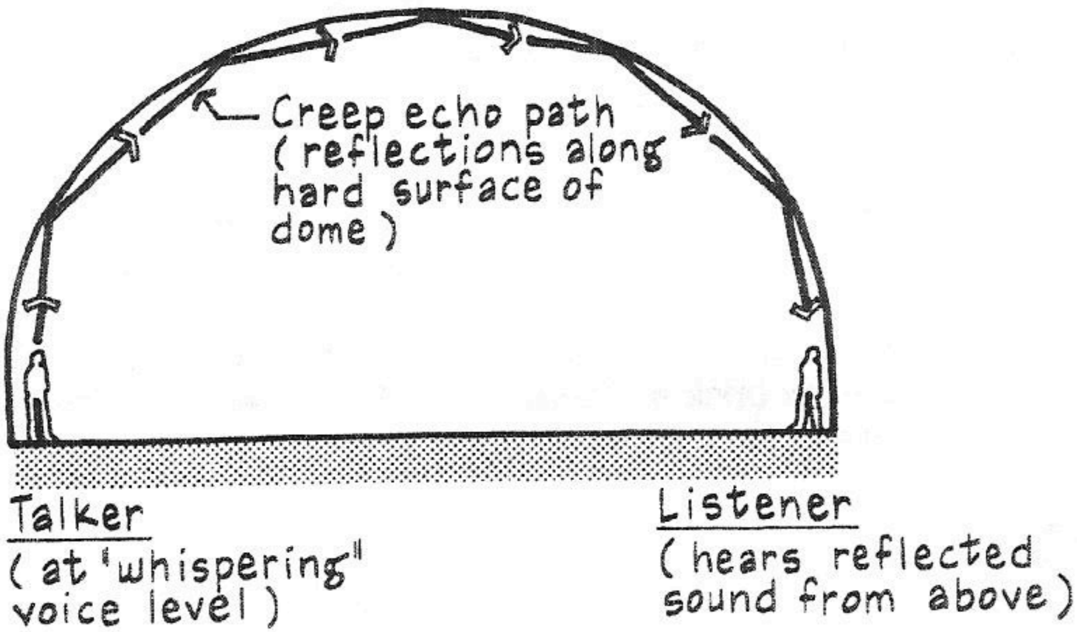
- Similar to a mirror reflection between two parallel mirrors. The images go on and on and on...
- Usually between two parallel surfaces, but can be between three or more
- Described as a clicking, ringing, buzzing or hissing
- Degrades speech intelligibility and produces tonal coloration of music

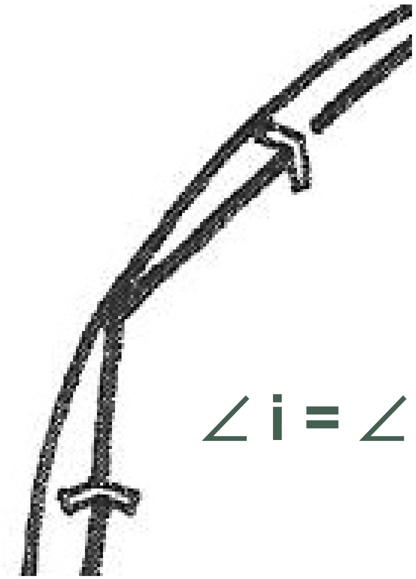


Small room with concave wall

Sound Reflection

Creep Echo



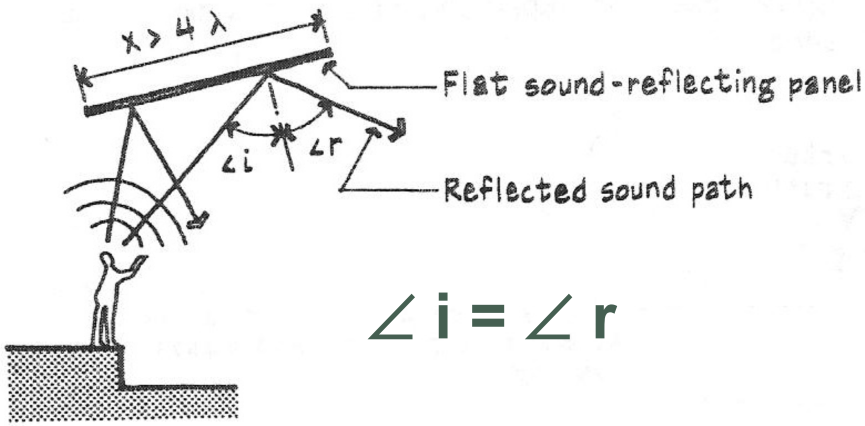


$\angle i = \angle r$

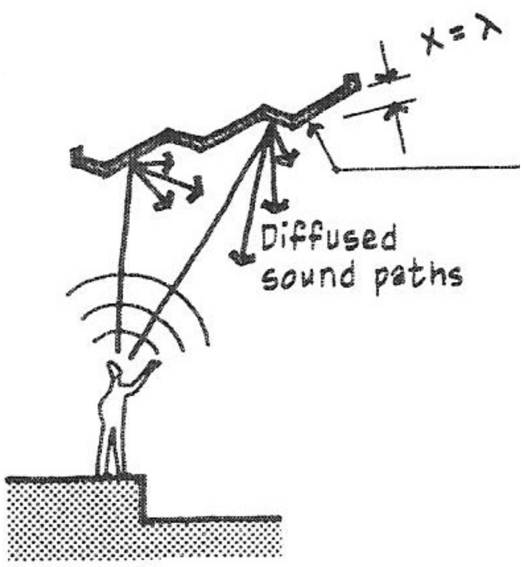
- Concave surface can have sound travel along its shape when the angle of incidence is shallow

Sound Diffusion

- **Specular reflection:** mirror type reflection where the angle of incidence equals the angle of reflection – "smooth" surface where irregularities are much smaller than a wavelength



- **Diffuse reflection:** incident sound is reflected in many directions – surface is heavily textured and irregularities are on the order of one wavelength for the sound to be diffusely reflected

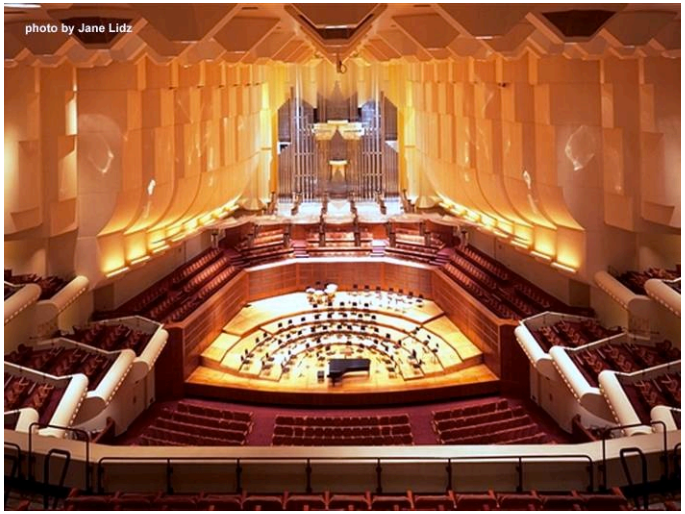


Sound Diffusion

- Irregular reflective room surfaces *increase* diffusion and absorptive surfaces *decrease* diffusion
- Interior ornamentation increases diffusion (ex., pilasters, piers, balconies, exposed beams, coffered ceilings, etc.)

Grosser Musikvereinssaal, Vienna

Photograph from www.musikverein.at

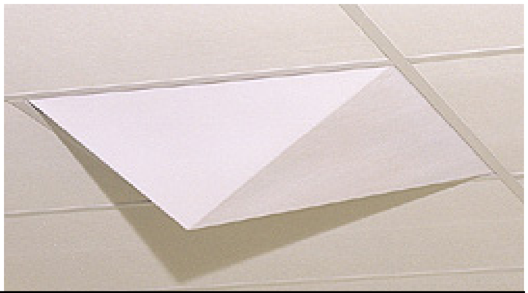


Louise M. Davies Symphony Hall, San Francisco

Photograph from www.kirkegaard.com

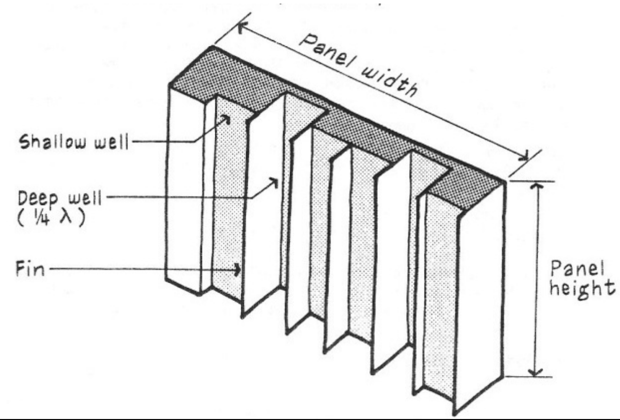
Sound Diffusion

Diffuser examples:



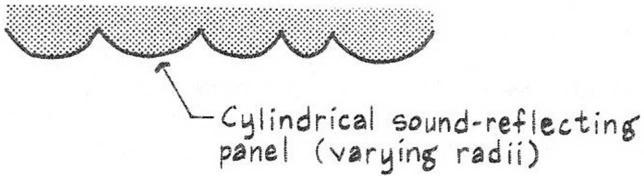
Kinetics Noise Control - Geometric Diffuser

RPG Diffusor Systems - Omnifusor



quadratic residue diffuser (QRD) – special diffuser designed to efficiently scatter sound

Convex Modulations

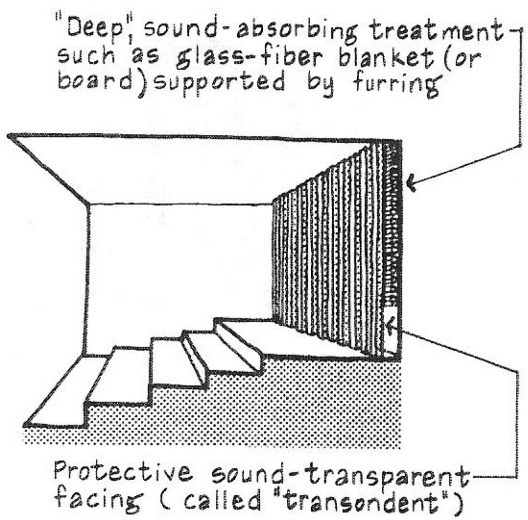


Wenger Corp. – wall diffuser panel

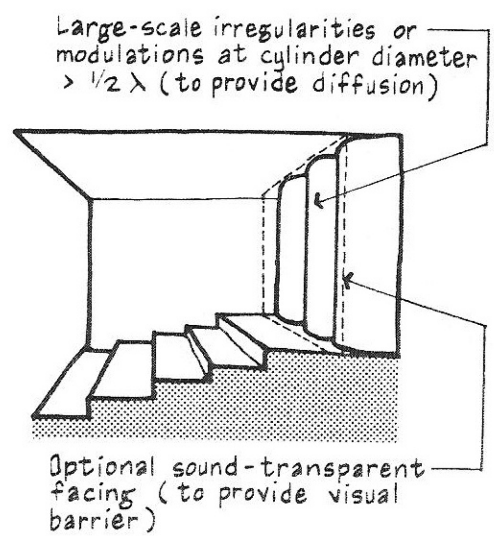
ECHO TREATMENT

Echo treatment involves reducing the reflection intensity via sound absorbing surfaces, diffusing the reflection or redirecting the reflection

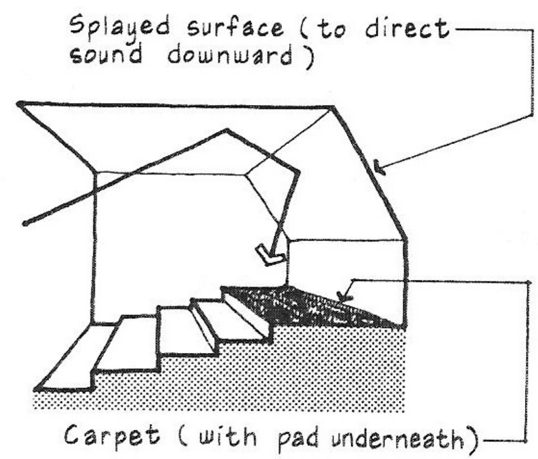
absorption



diffusion



reflection



Case Study

- Typical high school music rehearsal room
  - What acoustical treatments and design features do you see?

Case Study

- National Public Radio Studios
  - What acoustical treatments and design features do you see?

view from control room into studio

main studio

control room

Case Study

https://www.archdaily.com/889713/how-acoustic-shells-work-and-how-to-design-them-effectively/5a85fd87f197cc643c000004-how-acoustic-shells-work-and-how-to-design-them-effectively-photo

Reference

- M. David Egan, Architectural Acoustics (Acton, MA: Copley Custom Textbooks, 2008) ISBN 1-58152-507-9
- Mechanical & Electrical Equipment for Buildings (MEEB), 12th Edition, Walter T. Grondzik and Alison G. Kwok, John Wiley and Sons, Inc., 2015, ISBN: 978-1-118-61690-4